

IT694
Computer Networks
Lab - Routing Information Protocol

The Routing Information Protocol (RIP) is one of the oldest distance-vector routing protocols which employs the hop count as a routing metric. RIP prevents routing loops by implementing a limit on the number of hops allowed in a path from source to destination. The largest number of hops allowed for RIP is 15, which limits the size of networks that RIP can support. RIP implements the split horizon, route poisoning, and holddown mechanisms to prevent incorrect routing information from being propagated.

In RIPv1 routers broadcast updates with their routing table every 30 seconds. In the early deployments, routing tables were small enough that the traffic was not significant. As networks grew in size, however, it became evident there could be a massive traffic burst every 30 seconds, even if the routers had been initialized at random times. In most networking environments, RIP is not the preferred choice of routing protocol, as its time to converge and scalability are poor. However, it is easy to configure, because RIP does not require any parameters, unlike other protocols. RIP uses the User Datagram Protocol (UDP) as its transport protocol, and is assigned the reserved port number.

The Topology Diagram is shown in figure 1.

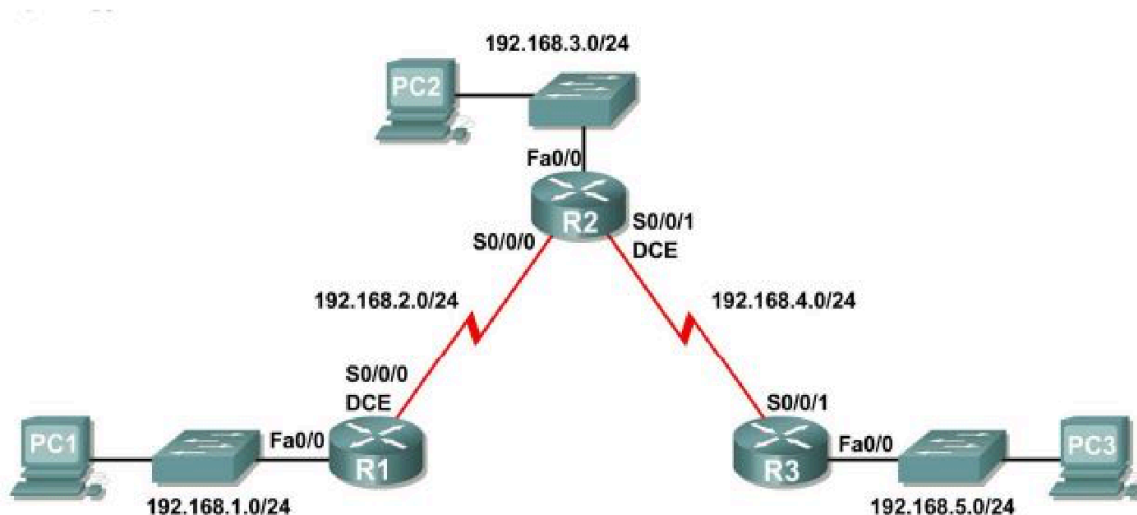


Figure 1: Topology to implement

The Reference Address Table is shown below

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	Fa0/0	192.168.1.1	255.255.255.0	N/A
	S0/0/0	192.168.2.1	255.255.255.0	N/A
R2	Fa0/0	192.168.3.1	255.255.255.0	N/A
	S0/0/0	192.168.2.2	255.255.255.0	N/A
	S0/0/1	192.168.4.2	255.255.255.0	N/A
R3	Fa0/0	192.168.5.1	255.255.255.0	N/A
	S0/0/1	192.168.4.1	255.255.255.0	N/A
PC1	NIC	192.168.1.10	255.255.255.0	192.168.1.1
PC2	NIC	192.168.3.10	255.255.255.0	192.168.3.1
PC3	NIC	192.168.5.10	255.255.255.0	192.168.5.1

1.1 Exercise

1. On the routers, enter global configuration mode and configure the hostname as shown on the chart. Then configure the console, virtual terminal lines password (both “cisco”) and privileged EXEC password (“class”):

2. Add the logging synchronous command to the console and virtual terminal lines: Router(config-line)#logging synchronous

Hints :

in CLI tab of the router:

<name> here <> is used for placeholders for name not actual name

HOW to enter global configuration mode and configure:

```
enable
configure terminal
hostname <router-name>
```

```
line con 0
```

```
password <password>
```

login

logging synchronous

HOW to configure console

line vty 0 4

password <password>

login

logging synchronous

enable secret <EXEC-password>

3. Disable DNS Lookup using "no ip domain-lookup" in router config.
4. Configure the interfaces on the R1, R2, and R3 routers with the IP addresses from the table under the Topology Diagram.
5. Use the "show ip interface brief" command to verify that the IP addressing is correct and that the interfaces are active.
6. Configure the Ethernet interfaces of PC1, PC2, and PC3 with the IP addresses and default gateways from the table under the Topology Diagram.
7. Test the PC configuration by pinging the default gateway from the PC.

1.2 Exercise

1. To enable RIP, enter the command "router rip" in global configuration mode.
2. Once you are in routing configuration mode, enter the classful network address for each directly connected network, using the network command with the following syntax:

Router(config-router)#network <network nr>

Router(config-router)#network <network nr>

3. Use the "show ip route" command to verify that each router has all of the networks in the topology entered in the routing table. Routes learned through RIP are coded with an R in the routing table.
4. Use the "show ip protocols" command to view information about the routing processes.
5. Use the "debug ip rip" command to view the RIP messages being sent and received.
6. Discontinue the debug output with the "undebug all" command.
7. Enter "show ip route rip" command.
8. Check the routing table for a specific route. (Ex: show ip route 192.168.1.0)
9. Configure your router to stop sending updates out the FastEthernet0/0 interface.
The "passive-interface fastethernet 0/0" command in routing configuration mode is used to disable sending RIPv1 updates out that interface.
10. Use show ip protocols to ensure FastEthernet0/0 is passive. You can also use debug ip rip to ensure that the router is not sending any RIP updates to FastEthernet0/0.

Note: Sometimes it is necessary to clear the dynamic routing table. Try the command

Router# clear ip route * on both R1 and R2 (you can also do it on the R3). This command will cause the routers to immediately flush routes in the routing table and request updates from each other.

"clear ip route *"

1.3 Questions

1. Display the output generated step 5 and what is the maximum number of hops seen in the output?
2. What is the administrative distance of RIP in Step 7?
3. When was the last update you saw in Step 8?
4. What is the default update time for RIP in Step 8?

Submission :

you are supposed to submit the answers of exercise 1.3 .

submit in the group and enter student ids of all group members.

The deadline for submission is 11:59 pm today.